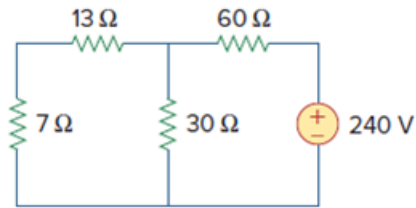


Problem

Determine the Thevenin equivalent circuit, shown in Fig. 4.101, as seen by the 7-ohm resistor. Then calculate the current flowing through the 7-ohm resistor.

Figure. 4.101:



Step-by-step solution

Step 1 of 6

Refer to Figure 4.101 in the text book.

Disconnect the $7\ \Omega$ resistor to calculate the value of the Thevenin's Voltage as shown in Figure 1.

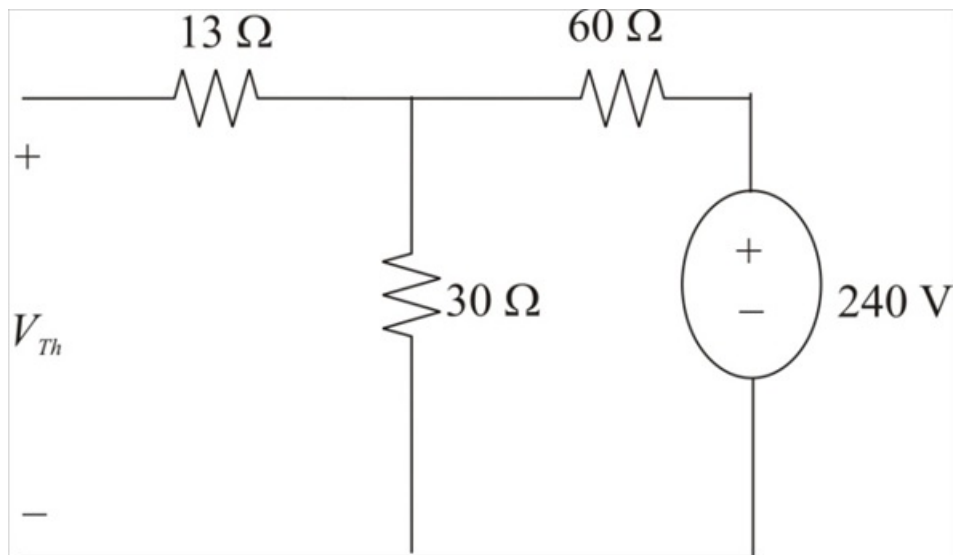


Figure 1

Step 2 of 6

Calculate the value of the voltage V_{Th} using voltage division rule.

$$\begin{aligned}
 V_{Th} &= (240) \left(\frac{30}{30 + 60} \right) \\
 &= (240) \left(\frac{30}{90} \right) \\
 &= (240) \left(\frac{1}{3} \right) \\
 &= 80\text{ V}
 \end{aligned}$$

Therefore, the value of the Thevenin's Voltage V_{Th} is 80 V.

Step 3 of 6

Deactivate the source as shown in Figure 2 to calculate the Thevenin's resistance.

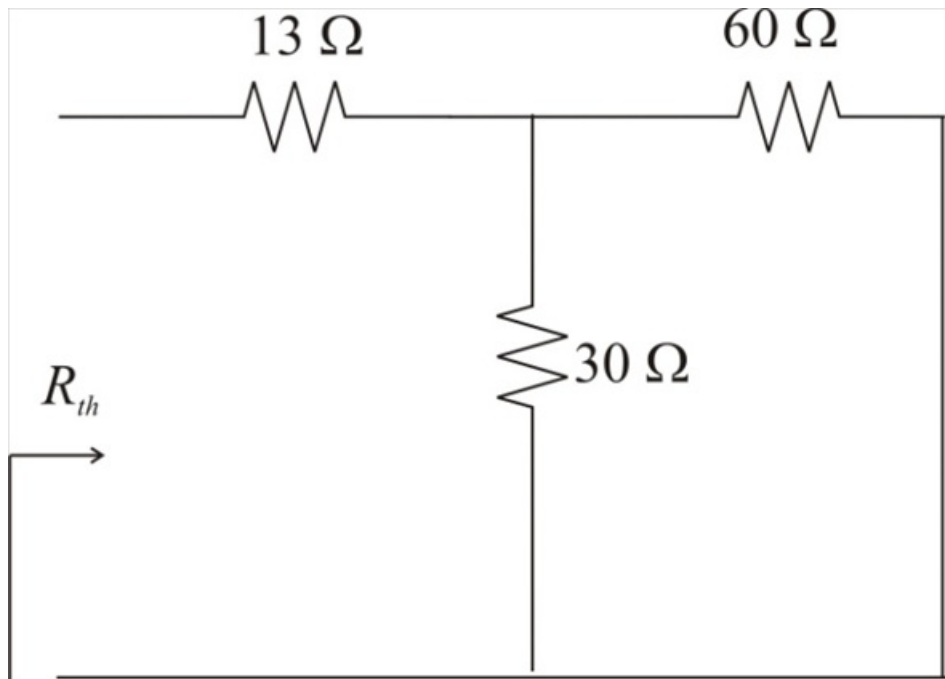


Figure 2

Step 4 of 6

Calculate the value of the Thevenin's resistance, R_{Th} .

$$\begin{aligned} R_{Th} &= 13 + (30 \parallel 60) \\ &= 13 + \frac{(30)(60)}{30 + 60} \\ &= 13 + \frac{1800}{90} \\ &= 13 + 20 \end{aligned}$$

$$R_{Th} = 33\ \Omega$$

Therefore, the value of the Thevenin's resistance R_{Th} is $33\ \Omega$.

Step 5 of 6

Draw the Thevenin's equivalent circuit diagram including the load resistance as shown in Figure 3.

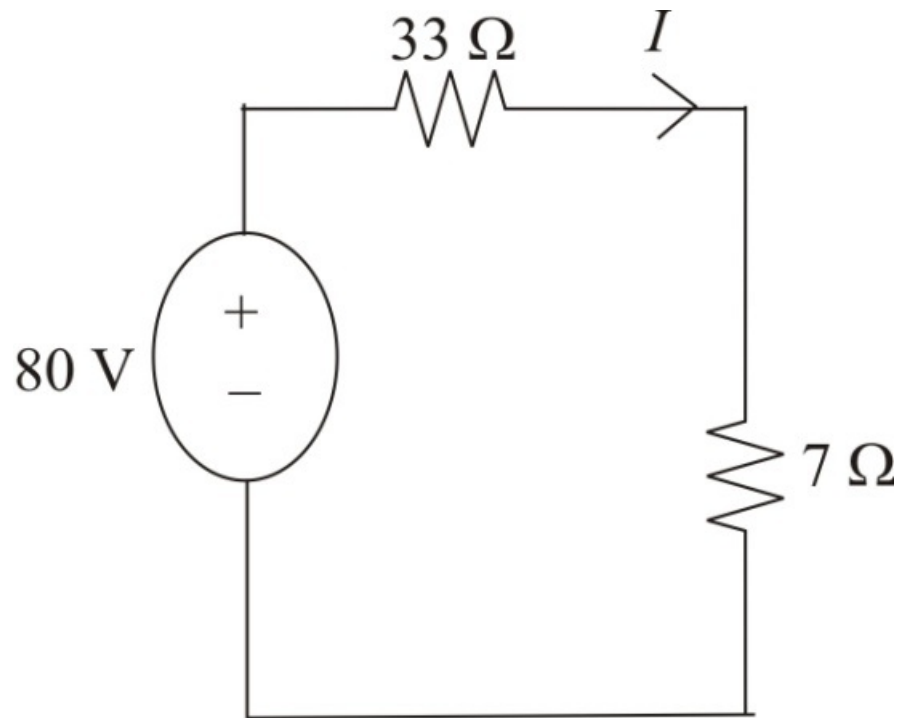


Figure 3

Step 6 of 6

Calculate the value of the current, I .

$$\begin{aligned} I &= \frac{80}{33+7} \\ &= \frac{80}{40} \\ &= 2 \text{ A} \end{aligned}$$

Therefore, the value of the current flowing through 7Ω resistance is 2 A.